

Chapter 6: Anatomy of Flowering Plants

Dicot Leaf

BAIO LEARNING SYSTEM · Narendra Sir — Founder/CEO

Class Notes · Topic 6/8

Dorsiventral (bifacial) leaf

Adaxial (upper) & abaxial (lower) epidermis, both with cuticle

More stomata on abaxial surface

Mesophyll differentiated: palisade + spongy parenchyma

Vascular bundle conjoint & closed, with bundle sheath

Epidermis

A dicot (dorsiventral) leaf has two epidermal layers: the upper surface is covered by the **adaxial epidermis** and the lower surface by the **abaxial epidermis**. Both bear a conspicuous waxy **cuticle**. The abaxial (lower) epidermis generally bears **more stomata** than the adaxial (upper) epidermis, which may even lack stomata.

Mesophyll

The tissue between the upper and lower epidermis is the **mesophyll**, made of parenchyma cells containing chloroplasts that carry out photosynthesis. Mesophyll has two types of cells: **palisade parenchyma** — elongated cells arranged vertically and parallel to each other, just below the adaxial epidermis; and **spongy parenchyma** — oval or round, loosely arranged cells below the palisade layer, extending to the lower epidermis, with numerous large air spaces and air cavities between them.

Vascular Bundle

Seen in the veins and the midrib. Surrounded by a layer of parenchymatous cells called the **bundle sheath**. The vascular bundle is **conjoint and closed**.

Dicot Leaf (Dorsiventral) – Key Regions

Region	Key Feature
Epidermis	Adaxial + abaxial, cuticle both sides, more stomata abaxially
Mesophyll	Differentiated — palisade (upper) + spongy (lower)
Vascular bundle	In veins/midrib, bundle sheath, conjoint & closed

Dorsiventral Leaf, 3 Regions

- Epidermis (adaxial & abaxial)
- Mesophyll (palisade + spongy)
- Vascular bundle (with bundle sheath)

